Class 8: Machine Learning Mini Project

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In today's mini-project we will explore a complete analysis using the unsupervised learning techniques covered in class (clustering and PCA for now).

The data itself comes from the Wisconsin Breast Cancer Diagnostic Data Set FNA breast biopsy data.

Data import

```r
wisc.df <- read.csv("WisconsinCancer.csv", row.names=1)
head(wisc.df)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
84358402         M       20.29        14.34         135.10    1297.0
843786           M       12.45        15.70          82.57     477.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
84358402         0.10030          0.13280         0.1980             0.10430
843786           0.12780          0.17000         0.1578             0.08089
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
84358402        0.1809                0.05883    0.7572     0.7813        5.438
```

```
843786          0.2087                0.07613    0.3345     0.8902        2.217
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
84358402   94.44      0.011490        0.02461      0.05688           0.01885
843786     27.19      0.007510        0.03345      0.03672           0.01137
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
84358402     0.01756             0.005115        22.54         16.67
843786       0.02165             0.005082        15.47         23.75
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
84358402          152.20     1575.0           0.1374            0.2050
843786            103.40      741.6           0.1791            0.5249
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
84358402          0.4000               0.1625         0.2364
843786            0.5355               0.1741         0.3985
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
84358402                 0.07678
843786                   0.12440
```

Remove the diagnosis column and keep it in a separate vector for later.

```r
diagnosis <- as.factor(wisc.df[,1])
wisc.data <- wisc.df[,-1]
head(wisc.data)
```

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean
842302	17.99	10.38	122.80	1001.0	0.11840
842517	20.57	17.77	132.90	1326.0	0.08474
84300903	19.69	21.25	130.00	1203.0	0.10960
84348301	11.42	20.38	77.58	386.1	0.14250
84358402	20.29	14.34	135.10	1297.0	0.10030
843786	12.45	15.70	82.57	477.1	0.12780

	compactness_mean	concavity_mean	concave.points_mean	symmetry_mean
842302	0.27760	0.3001	0.14710	0.2419
842517	0.07864	0.0869	0.07017	0.1812
84300903	0.15990	0.1974	0.12790	0.2069
84348301	0.28390	0.2414	0.10520	0.2597
84358402	0.13280	0.1980	0.10430	0.1809
843786	0.17000	0.1578	0.08089	0.2087

	fractal_dimension_mean	radius_se	texture_se	perimeter_se	area_se
842302	0.07871	1.0950	0.9053	8.589	153.40
842517	0.05667	0.5435	0.7339	3.398	74.08
84300903	0.05999	0.7456	0.7869	4.585	94.03
84348301	0.09744	0.4956	1.1560	3.445	27.23
84358402	0.05883	0.7572	0.7813	5.438	94.44
843786	0.07613	0.3345	0.8902	2.217	27.19

	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	0.006399	0.04904	0.05373	0.01587
842517	0.005225	0.01308	0.01860	0.01340
84300903	0.006150	0.04006	0.03832	0.02058
84348301	0.009110	0.07458	0.05661	0.01867
84358402	0.011490	0.02461	0.05688	0.01885
843786	0.007510	0.03345	0.03672	0.01137

	symmetry_se	fractal_dimension_se	radius_worst	texture_worst
842302	0.03003	0.006193	25.38	17.33
842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53
84348301	0.05963	0.009208	14.91	26.50
84358402	0.01756	0.005115	22.54	16.67
843786	0.02165	0.005082	15.47	23.75

	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.60	2019.0	0.1622	0.6656
842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
84358402	152.20	1575.0	0.1374	0.2050
843786	103.40	741.6	0.1791	0.5249

	concavity_worst	concave.points_worst	symmetry_worst

```
842302            0.7119                0.2654          0.4601
842517            0.2416                0.1860          0.2750
84300903          0.4504                0.2430          0.3613
84348301          0.6869                0.2575          0.6638
84358402          0.4000                0.1625          0.2364
843786            0.5355                0.1741          0.3985
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
84358402                 0.07678
843786                   0.12440
```

Exploratory data analysis

The first step of any data analysis, unsupervised or supervised, is to familiarize yourself with the data.

Q1. How many observations are in this dataset?

```r
nrow(wisc.data)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```r
table(wisc.df$diagnosis)
```

```
  B   M 
357 212 
```

Q3. How many variables/features in the data are suffixed with _mean?

First find the column names

```r
colnames(wisc.data)
```

```
 [1] "radius_mean"             "texture_mean"
 [3] "perimeter_mean"          "area_mean"
 [5] "smoothness_mean"         "compactness_mean"
 [7] "concavity_mean"          "concave.points_mean"
 [9] "symmetry_mean"           "fractal_dimension_mean"
[11] "radius_se"               "texture_se"
[13] "perimeter_se"            "area_se"
[15] "smoothness_se"           "compactness_se"
[17] "concavity_se"            "concave.points_se"
[19] "symmetry_se"             "fractal_dimension_se"
[21] "radius_worst"            "texture_worst"
[23] "perimeter_worst"         "area_worst"
[25] "smoothness_worst"        "compactness_worst"
[27] "concavity_worst"         "concave.points_worst"
[29] "symmetry_worst"          "fractal_dimension_worst"
```

Next I need to search within the column names for "_mean" pattern. The `grep()` function might help here.

```r
inds <- grep( "_mean", colnames(wisc.data))
length(inds)
```

```
[1] 10
```

Q. How many dimensions are in this dataset?

```r
ncol(wisc.data)
```

```
[1] 30
```

Principal Component Analysis

First do we need to scale the data before PCA or not.

```r
round( apply(wisc.data, 2, sd), 3)
```

```
          radius_mean           texture_mean         perimeter_mean
                3.524                  4.301                 24.299
            area_mean        smoothness_mean       compactness_mean
```

```
               351.914                   0.014                   0.053
        concavity_mean     concave.points_mean           symmetry_mean
                 0.080                   0.039                   0.027
 fractal_dimension_mean               radius_se              texture_se
                 0.007                   0.277                   0.552
          perimeter_se                 area_se           smoothness_se
                 2.022                  45.491                   0.003
        compactness_se            concavity_se       concave.points_se
                 0.018                   0.030                   0.006
           symmetry_se    fractal_dimension_se            radius_worst
                 0.008                   0.003                   4.833
         texture_worst         perimeter_worst              area_worst
                 6.146                  33.603                 569.357
      smoothness_worst       compactness_worst         concavity_worst
                 0.023                   0.157                   0.209
   concave.points_worst          symmetry_worst fractal_dimension_worst
                 0.066                   0.062                   0.018
```

Looks like wee need to scale.

```r
# Perform PCA on wisc.data by completing the following code
wisc.pr <- prcomp( wisc.data, scale=TRUE )
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
```

```
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal components (PC1)?

44.27%

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

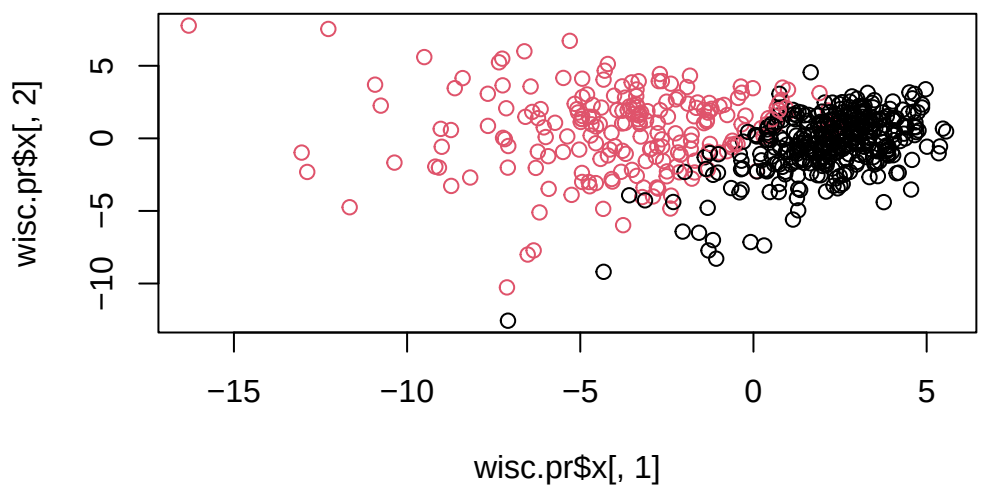
3 PCs capture 72%

> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

PC plot

We need to make our plot of PC1 vs PC2 (a.k.a score plot, PC-plot, etc.). The main result of PCA…

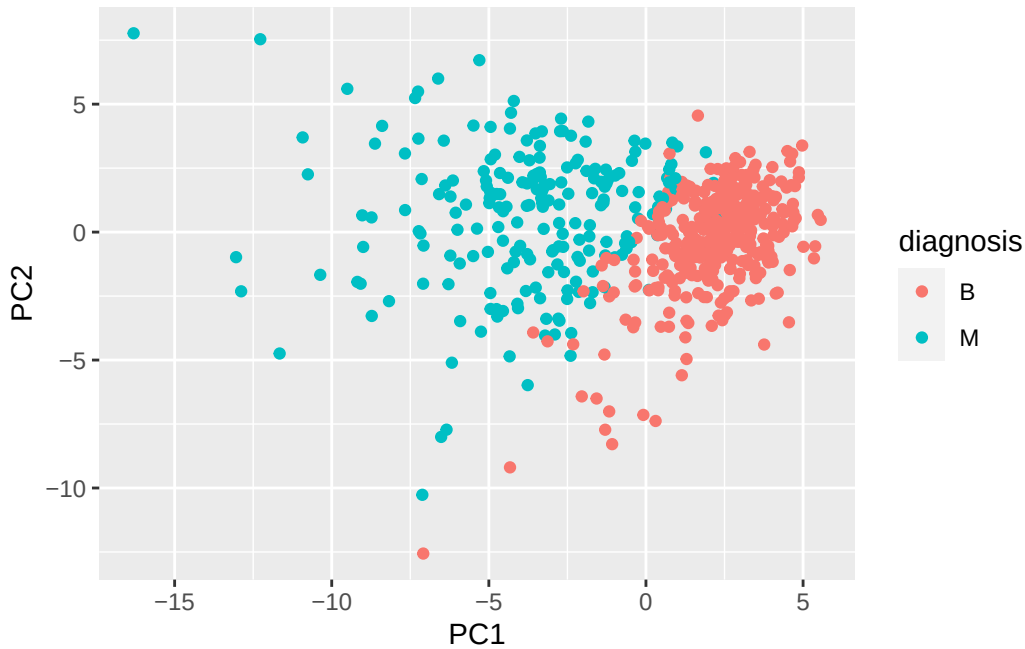
```r
plot(wisc.pr$x[,1], wisc.pr$x[,2], col=diagnosis)
```



```r
library(ggplot2)

pc <- as.data.frame(wisc.pr$x)
pc$diagnosis <- diagnosis

ggplot(pc) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Variance explained

We can get this from the outupt of the `summary()` function.

```r
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
```

```
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

Calculate the variance of each principal component by squaring the sdev component of wisc.pr (i.e. wisc.pr$sdev^2). Save the result as an object called pr.var.

```r
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

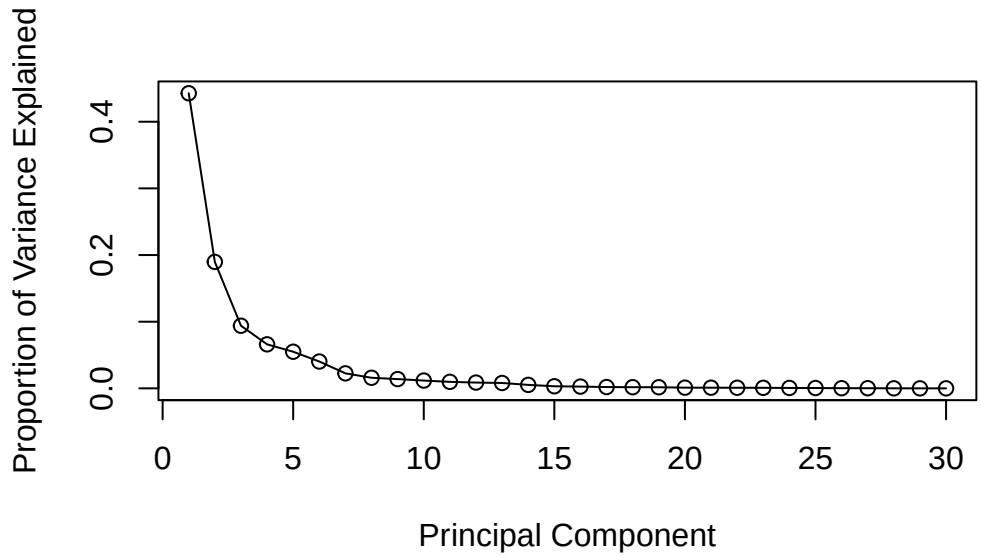
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

Calculate the variance explained by each principal component by dividing by the total variance explained of all principal components.

```r
pve <- pr.var / sum(pr.var)
head(pve)
```

```
[1] 0.44272026 0.18971182 0.09393163 0.06602135 0.05495768 0.04024522
```

```r
plot(pve, xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     type = "o")
```



Examine the PC loadings

How much do the original variables contribute to the new PCs that we have calculated? To get at this data we can look at the the `$rotation` component of the returned PCA object.

```
head(wisc.pr$rotation[,1:3])
```

```
                        PC1         PC2          PC3
radius_mean      -0.2189024  0.23385713 -0.008531243
texture_mean     -0.1037246  0.05970609  0.064549903
perimeter_mean   -0.2275373  0.21518136 -0.009314220
area_mean        -0.2209950  0.23107671  0.028699526
smoothness_mean  -0.1425897 -0.18611302 -0.104291904
compactness_mean -0.2392854 -0.15189161 -0.074091571
```

Focus in on PC1

```
head(wisc.pr$rotation[,1])
```

```
    radius_mean     texture_mean   perimeter_mean        area_mean
     -0.2189024       -0.1037246       -0.2275373       -0.2209950
smoothness_mean compactness_mean
     -0.1425897       -0.2392854
```

Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean?

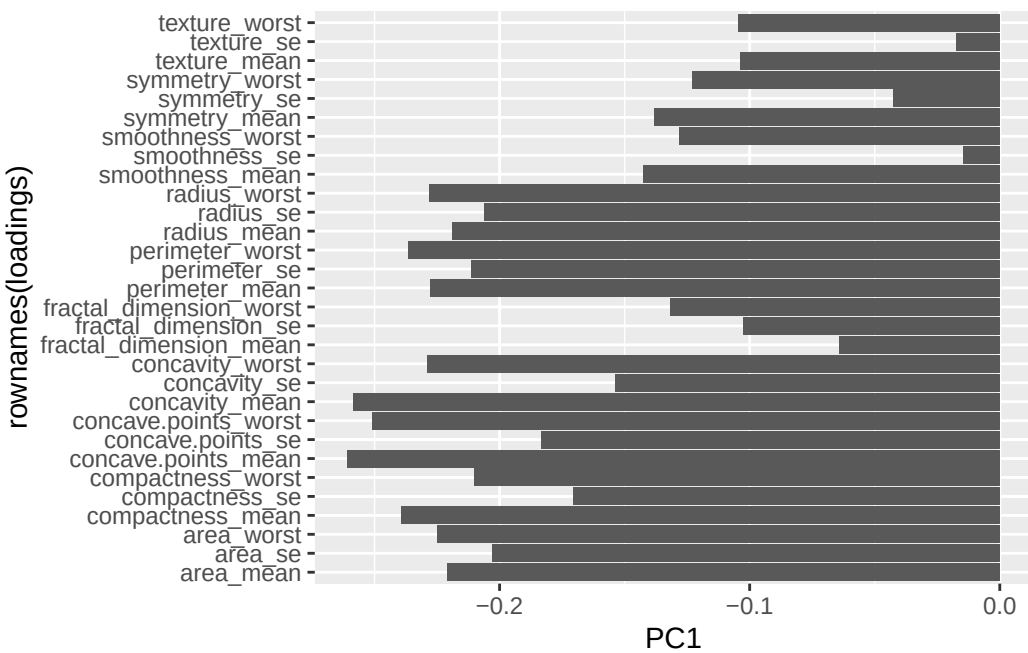
```r
wisc.pr$rotation["concave.points_mean" ,1]
```

```
[1] -0.2608538
```

There is a complicated mix of variables that go together to make up PC1 - i.e. there are many of the original variables that together contribute highly to PC1.

```r
loadings <- as.data.frame(wisc.pr$rotation)

ggplot(loadings) +
  aes(PC1, rownames(loadings)) +
  geom_col()
```



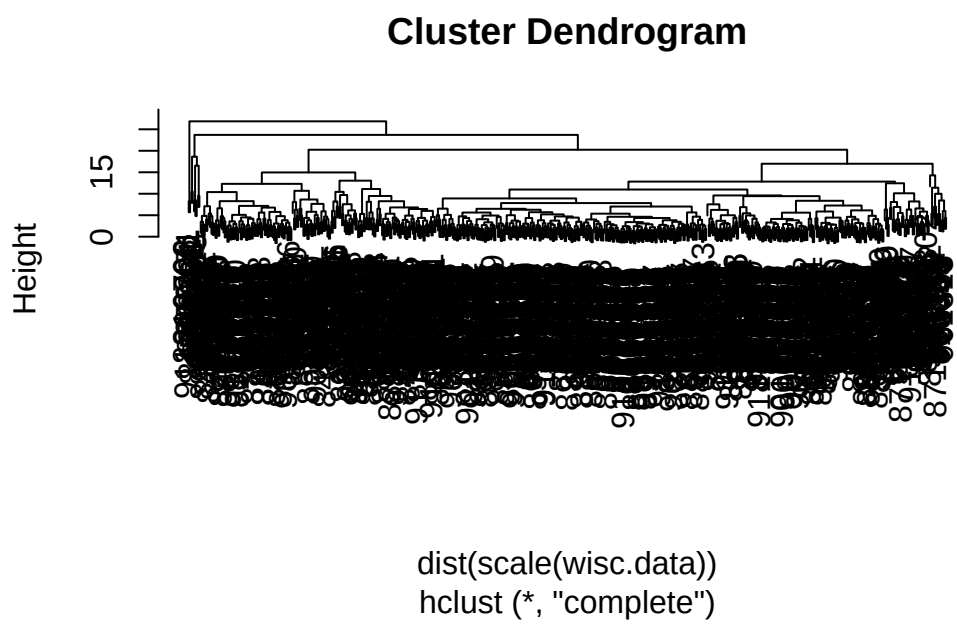
3. Hierarchical clustering

The goal of this section is to do hierarchical clustering of the original data.

First we will scale the data, then distance matrix, then hclust

```
wisc.hclust <-  hclust( dist( scale(wisc.data) ) )
```

```
plot(wisc.hclust)
```



Cut this tree to yield cluster membership vector with `cutree()` function.

```
grps <- cutree(wisc.hclust, h=19)
table(grps)
```

```
grps
  1   2   3   4
177   7 383   2
```

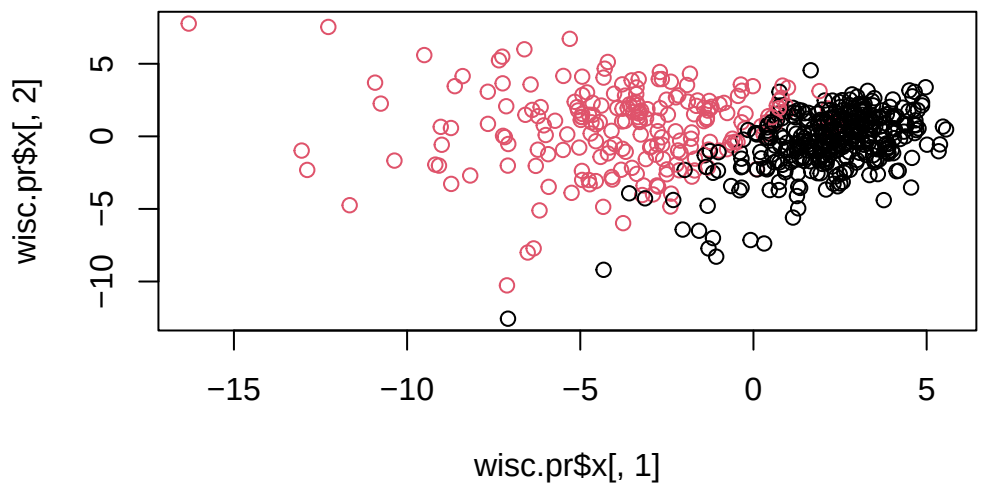
```
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  12 165
   2   2   5
   3 343  40
   4   0   2
```

Combine methods: PCA and HCLUST

My PCA results were intresting as they showed a seperation of M and B samples along PC1.

```
plot(wisc.pr$x[,1], wisc.pr$x[,2], col=diagnosis)
```



I want to cluster my PCA results - that is use `wisc.pr$x` as input to `hclust()`.

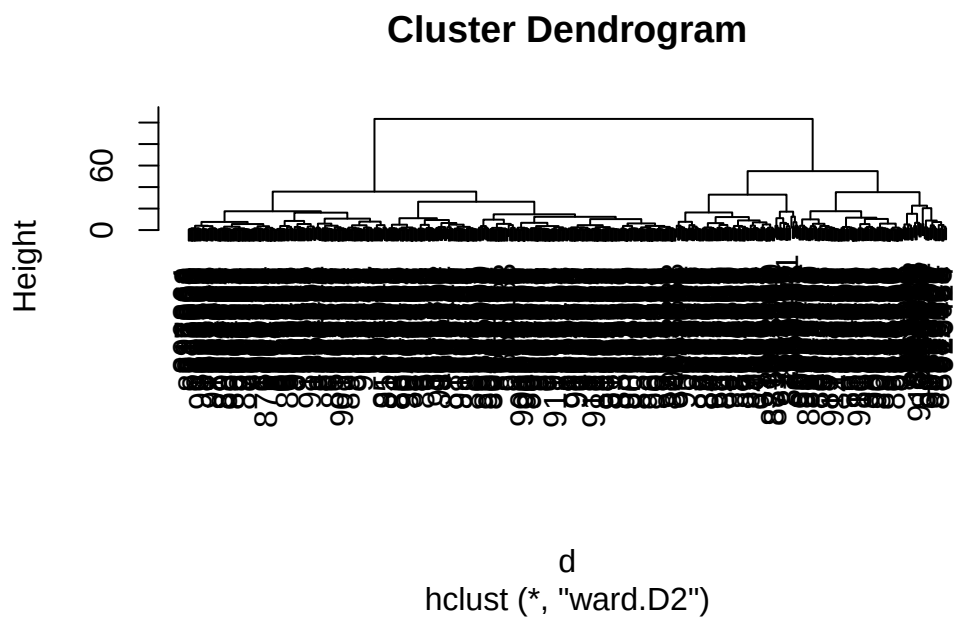
Try clustering in 3 PCs, that is PC1, PC2 and PC3 as input

```
d <- dist(wisc.pr$x[,1:3])

wisc.pr.hclust <- hclust(d, method="ward.D2")
```

An my tree result figure

```
plot(wisc.pr.hclust)
```

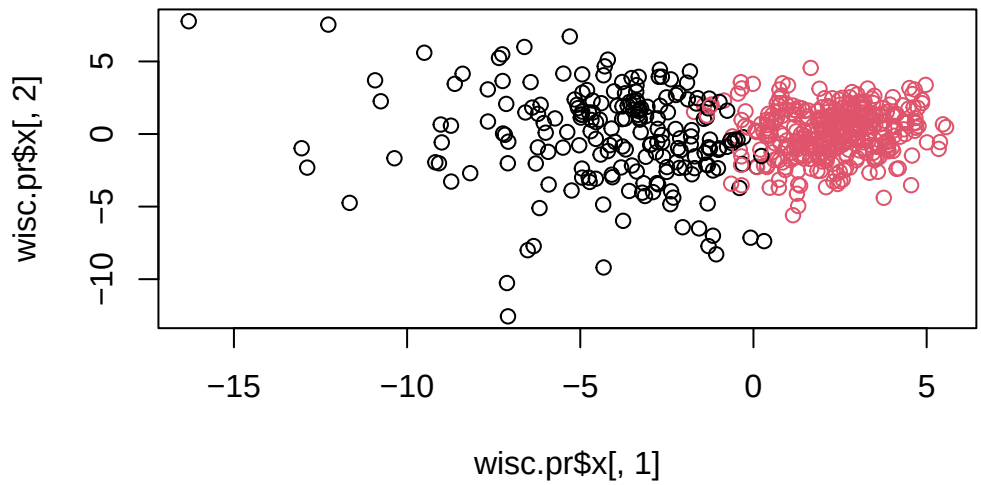


Let's cut this tree into two groups/clusters

```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1   2
203 366
```

```
plot(wisc.pr$x[,1], wisc.pr$x[,2], col=grps)
```



How well do the two clusters separate the M and B diagnosis?

```
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  24 179
   2 333  33
```

```
(179+333)/nrow(wisc.data)
```

```
[1] 0.8998243
```